Performance Comparison of Adaptive Filter Algorithms for Noise Cancellation

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Abstract—**The purpose of adaptive filter is to solve the dilemma of fast convergence rate and low mean square error. In the past two decades many adaptive filter algorithms have been presented and have claimed that they have good convergence speed and tracking properties. This paper describes the concept of adaptive noise cancellation, an alternative method of estimating signals corrupted by additive noise or interference. This paper summarizes three promising algorithms and gives a performance comparision via extensive simulation. Step size is the main parameter for the convergence rate and mean square error. The simulation analysis showed that RLS algorithm had faster convergence speed and smaller steady state error compared with basic LMS algorithm.The results shown, for convergence rate with less no of taps is the most difficult phase in signal processing applications for the perfect working of any system.**

Keywords—Adaptive filters; Least mean square; normalized least mean square; Recursive least square

I. INTRODUCTION

Adaptive filters have been used in many applications such as system identification, Channel equalization, Channel estimation, echo cancellation and active noise control[6]. Linear and nonlinear filtering techniques have received increasing attentions in recent adaptive signal processing literatures[6].Numerous researchers have contributed to the development in these fields[6]. Several adaptive filter algorithms are proposed for noise cancellation. The adaptive filter essentially minimizes the mean-squared error between noisy signal and a reference signal. In adaptive filters convergence rate is decreasing by increasing the number of taps especially if the reference signal spectrum has a large dynamic range [1]. Noise cancellation requires a method for adjusting the learning rate when noise or interference present in the signal. Most of the echo cancellation algorithms used to detect double talk conditions so have long been recognized as an essential component of two way voice communication systems for reducing the annoying effects of network and acoustic echoes [13].The least mean square (LMS) and normalized least–mean square (NLMS) algorithms are the most widely used adaptive filter algorithms because they are easy to implement and have low computational complexity[8]. The stability of this algorithm is governed by step size parameter. It is well known that the choice of this parameter, within the stability conditions, reflects a tradeoff between fast convergence and good tracking ability on one side and low misadjustment on the other hand.Although NLMS algorithm has achieved a certain degree of success so it converges slowly with colored input signals [8]. The rest of the paper is organized as follows. Section II provides the overview of different parameters, Section III discuss about the different filters, then in Section IV different adaptive algorithms are discussed, followed by simulation results and conclusion .

II. PARAMETER

The convergence rate, tracking, misadjustment, mean square error, tap length are the few parameters that we will discuss in this paper.The number of coefficients of a linear filter is a important parameter that effects the performance of a minimum mean square error (MMSE) of the adaptive filter [6]. MMSE is a monotonic non increasing function of the tap length but due to the increase in tap length, there is a decrease of MMSE. It doesn't increase complexity but introduces adaptation noise.So we use a optimum tap length that balances the steady state performance and complexity. In multiuser echo cancellation system the length of the echo may keep changing as user entering or leaving the system. but most of the designs tap length is usually fixed[6].

III. FILTERS

Filters are of Linear and non linear types that are used in signal processing applications. All the applications where linear and nonlinear distortions have to be identified and compensated by adaptive signal processing[1],[2]. FIR and IIR both are linear filters but Volterra is a non linear filter. By using these filters people are working on time/frequency domain implementation but few are working on statistical methods for higher order spectral analysis[6].

A. Wiener Filter Theory

It is a statistical analysis method where prior environment is known before the processing [1]. This theory is to define very clearly by mean of an optimum filter. The mean square error(MSE), ξ, is defined by the "expectation" of the squared error .

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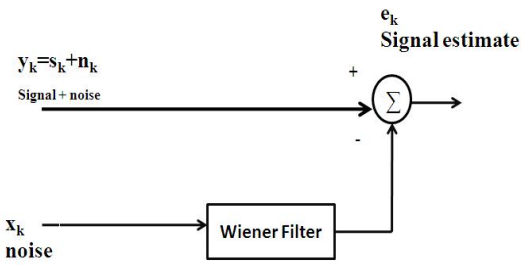


Fig 1: Weiner filter

$$e_k = y_k - \hat{n} = y_k - \sum_{i=0}^{N-1} w(i).x_{k-i} \quad (1)$$

$$\xi = E\{e_k^2\} = E\{y_k^2\} - 2W^T R_{yX} + W^T R_{XX} W \quad (2)$$

Weiner filter theory is defined by performance surface[1]-[2].

B. Performance Surface

The vertical axis represents the mean square error and the two horizontal axes represent the values of two filter coefficients [2].The quadratic error function, or performance surface, can be used to determine the optimum weight vector Wopt (or Wiener filter coefficients). With a quadratic performance function there is only one global optimum; no local minima exist. The shape of the function would be hyper-parabolic if there were more than two weights [2]. When the weight vector (filter coefficient) is at the optimum Wopt, the mean square error will be at its minimum. The filter given by Wopt is the Wiener filter. Wopt has to be calculated repeatedly for non-stationary signals and this can be computationally intensive. An iterative solution to the Wiener-hopf equation is the steepest decent algorithm[2].

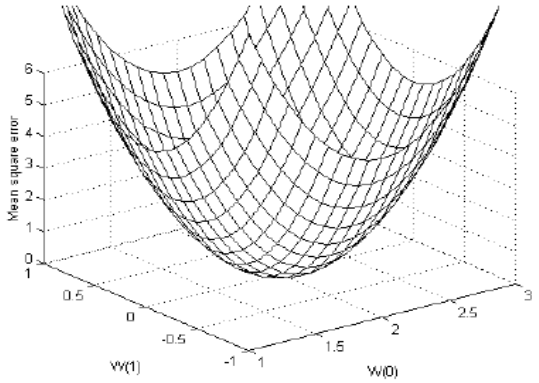


Fig 2: Performance surface

C. Adaptive Filter

It is ansteepest descent algorithm, the characteristics of systems are either not known or time variable. Therefore, processing approaches should adapt the unknown characteristics, hence to extract valid information in a changing scenario. For that reason the adaptive algorithms should be simple, computationally efficient, implementable on the existing hardware platform (like digital signal processors or configurable blocks) and cost effective for commercial utility. Any real time processing has to be adaptive and some of the common applications are: compression, coding, active control of noise inside aircraft cabin and automobiles, industrial noise and vibration, communication applications like channel equalization, acoustic and line echo cancellation, adaptive antenna arrays and adaptive processing of biomedical signals[1]-[2].

Adaptive filters have the ability to adjust their own parameters automatically and their design requires little or no a priori knowledge of signal or noise characteristics [6].The design of such filters is the domain of optimal filtering which originated with the wiener filter and was extended by work of kalman filter.

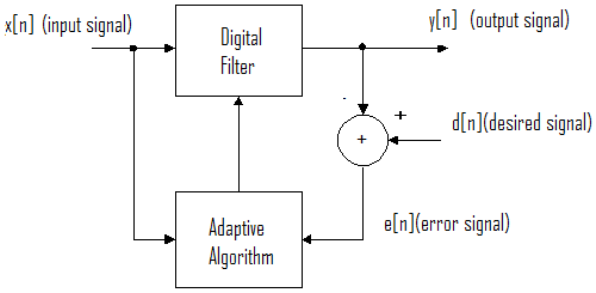


Fig. 3. Adaptive Filter

D. Volterra Filter

It can deal with general class of nonlinear systems but its output is still linear with respect to various higher order kernels or impulse responses [6]. So truncated Volterra models have been widely applied and became very popular. One key reason is that the number of the volterra coefficients increases geometrically as the delays and orders increasing [6].

IV. ADAPTIVE ALGORITHMS

Most popular adaptive filter algorithms are Least Mean square (LMS) algorithm and normalized Least Mean square (NLMS) algorithm [1]. The popularity of these algorithms is due to its simplicity and robust performance [2].

A. Least Mean Square (LMS) Algorithm

The LMS is one of the simplest algorithms used in the adaptive structures because of its simplicity, ease of implementation and low computational complexity.It uses error signal to calculate coefficients.In this diagram X(n) was the signal sequence, W(n) was the weight vector of the adaptive filter, it consisted of M weights of the adaptive

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filter,the output response was y(n), reference signal was d(n) and M was the order of the adaptive filter.

$$X(n)=[x(n),x(n-1),\ldots\ldots\ldots\ldots,x(n-M+1)] \quad (3)$$

$$W(n)=[w1,w2,\ldots\ldots\ldots\ldots..,wn]^T \quad (4)$$

No of taps of a linear filter is an important parameter for calculating the performance of minimum mean square error (MMSE). By adjusting the tap length we can improve convergence rate. Segmented filter (SF) was used for fixed tap length and gradient descent (GD) was used for variable tap length [2].

1.Filter output :

$$y(n)=w^H(n)X(n) \quad (5)$$

2.Estimation error :

$$e(n)=d(n)-W^H(n)X(n) \quad (6)$$

And the performance function of the filter was $\zeta^*(n)$ labeled as

$$\zeta(n)=\min(E[e^2(n)]) \quad (7)$$

3. Standard LMS function was

$$W(n+1)=W(n)+\mu X(n)e^*(n) \quad (8)$$

$$0< \mu \ \mu<1/\lambda max \quad (9)$$

Where μ was step size parameter,it determines the convergence rate [2].The λmax was the maximum eigen value of the autocorrelative matrix of input signal,but it was unknown in practice.Where e*(n) denotes the complex conjugate of error signal e(n).

B. Normalized Least Mean Square (NLMS) Algorithm

NLMS is widely used algorithm because of its simplicity and robust performance. The stability of the basic NLMS is controlled by a step size[6]. This parameter also governs the rate of convergence, speed, tracking ability and the amount of steady-state excess mean-square error (MSE) [6]. Aiming to solve conflicting objectives of fast convergence and low excess MSE, it converges slowly with colored input signals [1]-[2].Rate of convergence of NLMS is faster than standard LMS algorithm for both uncorrelated and correlated input data.

In the standard LMS algorithm if x(n) is large, it suffers from gradient noise amplification[2]. But normalized LMS algorithm seeks to avoid gradient noise amplification [2]. And sequence of events is repeated for a number of iterations until filter reaches asteady state [2].The step size is time varying m (n), and optimized to minimize error.

$$\hat{w}^H(n+1)u(n)=d(n) \quad (10)$$

$$\hat{w}(n+1)=\hat{w}(n)+\mu/(\delta+\square u(n)\square^2)u(n)e^*(n) \quad (11)$$

$$W(n)+\mu(n)[p-RW(n)] \quad (12)$$

C. RLS Algorithm

Recursive algorithm is used for the design of adaptive transversal filters which provides the least square estimate of the tap weight vector of the filter at iteration (n-1) and also compute the updated estimate of the vector at the iteration n upon the arrival of new data [2]. Rate of convergence is typically an order of magnitude faster than that of simple LMS filter. so RLS filter whitens the input data by using inverse correlation matrix of the data with zero mean,but this improvement in performance increase the computational complexity of the RLS filter [2]. RLS algorithms calculate J(n) by using the following equation:

$$J(n)=1/N\Sigma\lambda^i e^2(n-1) \quad (13)$$

In Equation (11) where N is the filter length and λ is the forgetting factor. This algorithm calculates not only the instantaneous value e2(n) but also the past values, such as e2(n–1), e2(n–2), ..., e2(n–N+1). The range of the forgetting factor is (0, 1). When the forgetting factor is less than 1, this factor specifies that this algorithm places a larger weight on the current value and a smaller weight on the past values [2]. The resulting E[e2(n)] of the RLS algorithms is more accurate than that of the LMS algorithms[2].The LMS algorithm require fewer computational requirements and memory as compare to RLS algorithm. However, the eigenvalue of the input correlation matrix, or the correlation matrix of the input signal, might affect the convergence speed of the resulting adaptive filter[2]. The convergence speed of the RLS algorithms is much faster than that of the LMS algorithms. However, the RLS algorithms require more computational resources than the LMS algorithms.

$$E(i)=d(i)-y(i) \quad (14)$$

$$=d(i)-w^H(n)u(i) \quad (15)$$

$$W(n)=w(n+1)+k(n)\zeta^*(n) \quad (16)$$

V. SIMULATION RESULTS

Several experimental results will be shown in this section in order to prove the effectiveness of three algorithms, focusing on the two main aspects i.e., convergence rate and MSE.In this section we present simulation results with different values of μ=0.2,0.008,1,10 with input signal is random signal and Noise is also random signal with different amplitude.

Aiming to solve the conflicting objectives of fast convergence and low excess MSE associated with the conventional LMS, a number of adaptive filter algorithms have been presented in the table.We compare the LMS , NLMS and RLS algorithm as a noise canceller. Smaller step size with slow convergence rate and larger step size provides fast convergence rate. Figure 4 shows the comparision between three algorithms for noise cancellation and figure 5 to 7 shows the comparison between LMS, NLMS and RLS with different values of taps and calculate

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convergence rate. And figure 8 to 10 shows MSE Vs Time index for different values of μ.Figure 11 shows the comparion between LMS and NLMS for mean square error. This LMS employs a larger step size when the estimation error is large, and vice versa. Extensive simulations show that the steady-state behavior predicted by the analysis is in very good agreement with the experimental results.

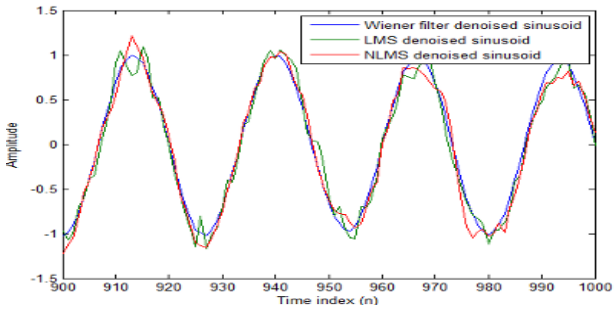
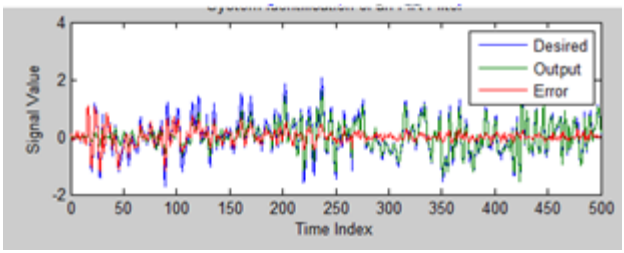


Fig 4: Comparision between Weiner , LMS and NLMS for output

LMS with μ=0.008



LMS with μ =0.2

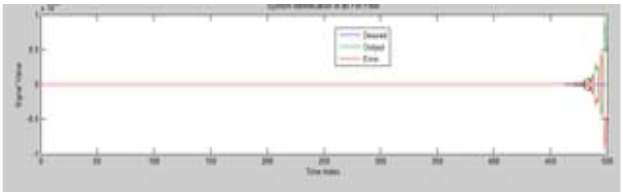
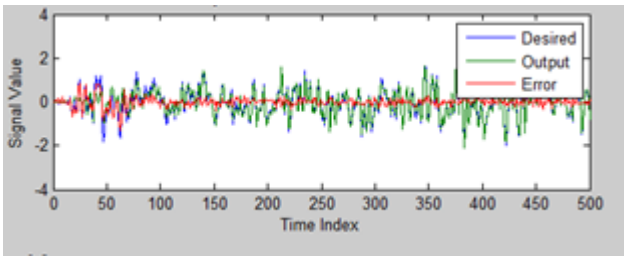


Fig 5: Convergence curve with LMS

NLMS with μ =1



NLMS with μ =4

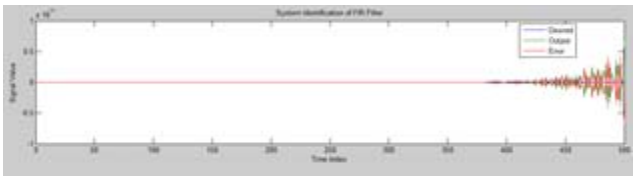
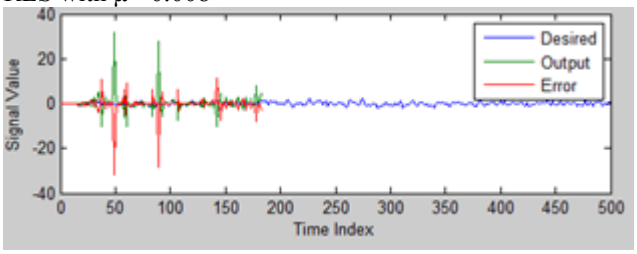


Fig 6: Convergence curve with NLMS

RLS with μ =0.008



RLS with μ =0.001

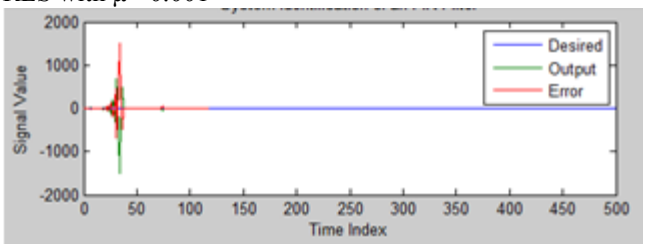


Fig 7: Convergence curve with R LS

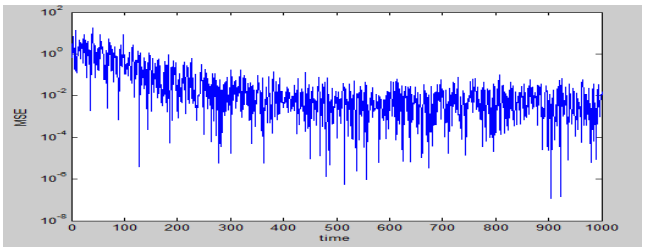


Fig 8: MSE VS Time for LMS with μ =0.005

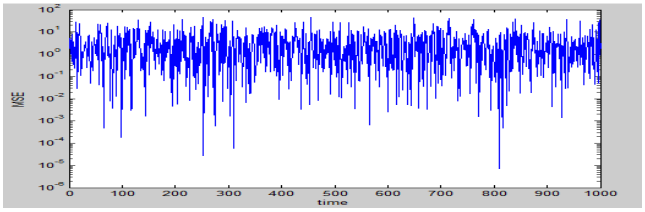


Fig 9: MSE VS Time for LMS with μ =0.0005

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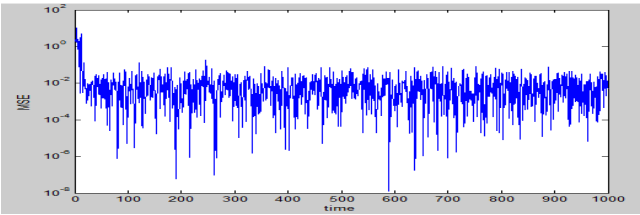


Fig 10: MSE VS Time for LMS with μ =0.05

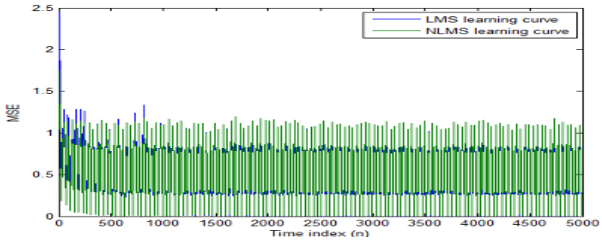


Fig 11: Comarision between LMS and NLMS

TABLE I. COMPARISON TABLE

Algorithms	Step size	Convergence Rate	MSE	Computational Complexity	Stability
LMS	Low(variable)	Low	High	Low	Less
NLMS	Controlled	Faster	Low	Low	More
RLS	Variable	Fastest	Low	more	More

VI. CONCLUSION

Besides the LMS algorithm, there are other algorithms like RLS and kalman , more powerful with higher convergenc speed.Three adaptive filter algorithms have been compared by simulation to achieve high convergence rate and minimum mean square error with noise and different values of μ. Every algorithm works on different methods for noise cancellation and improves system performance. But in all these algorithms authors worked on different parameters and try to improve their results.In future we want to implement these algorithms for real time signals like recorded speech with different background noise.

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